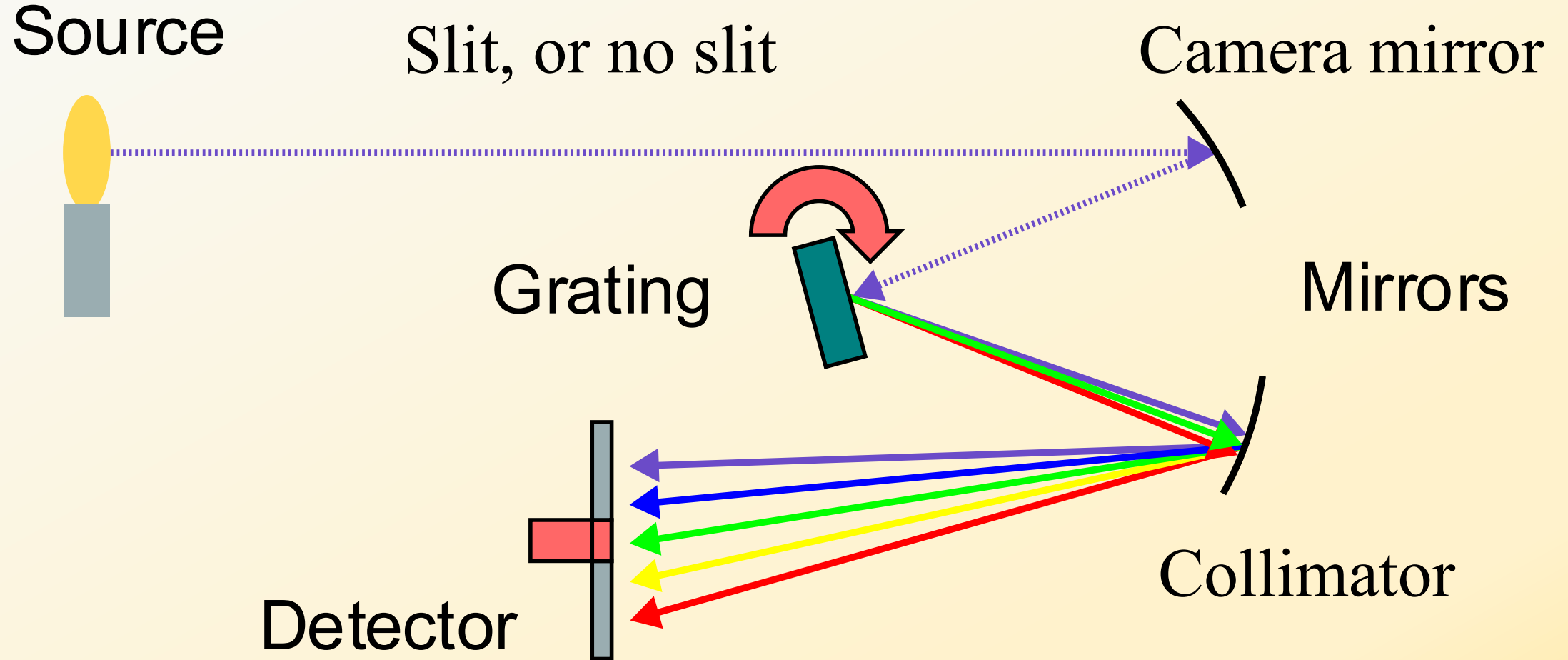
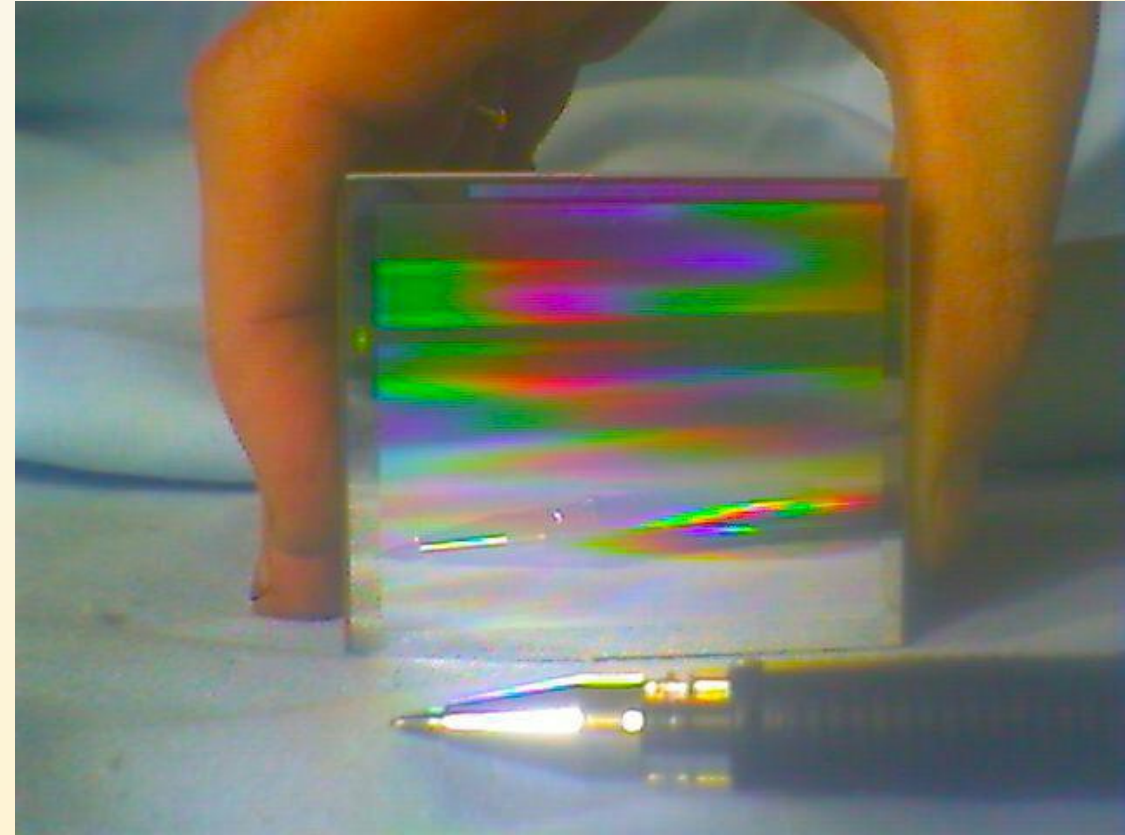
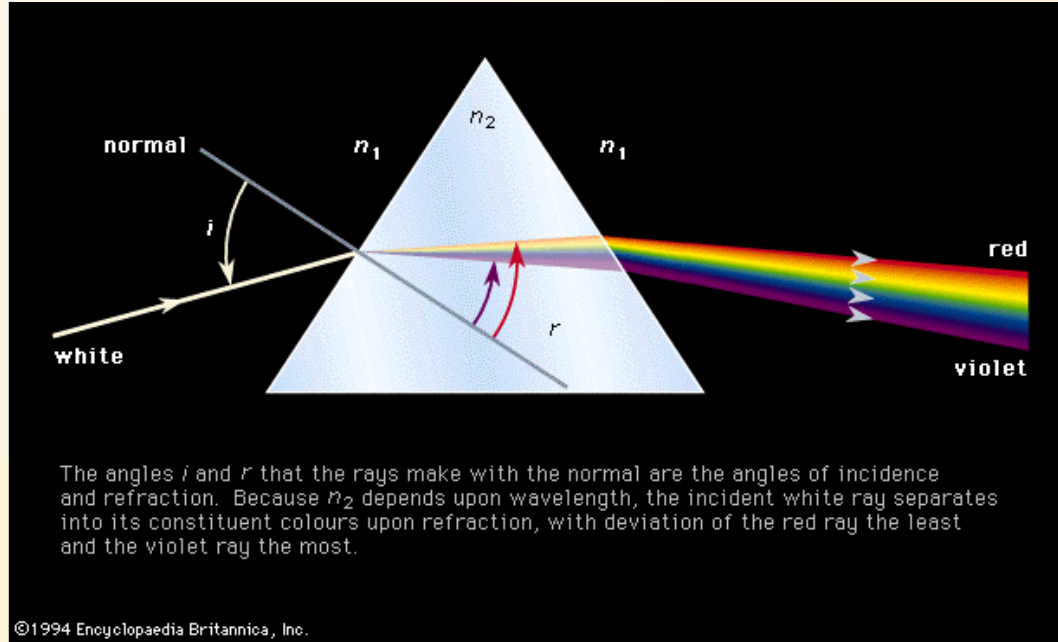


The properties of spectroscopes



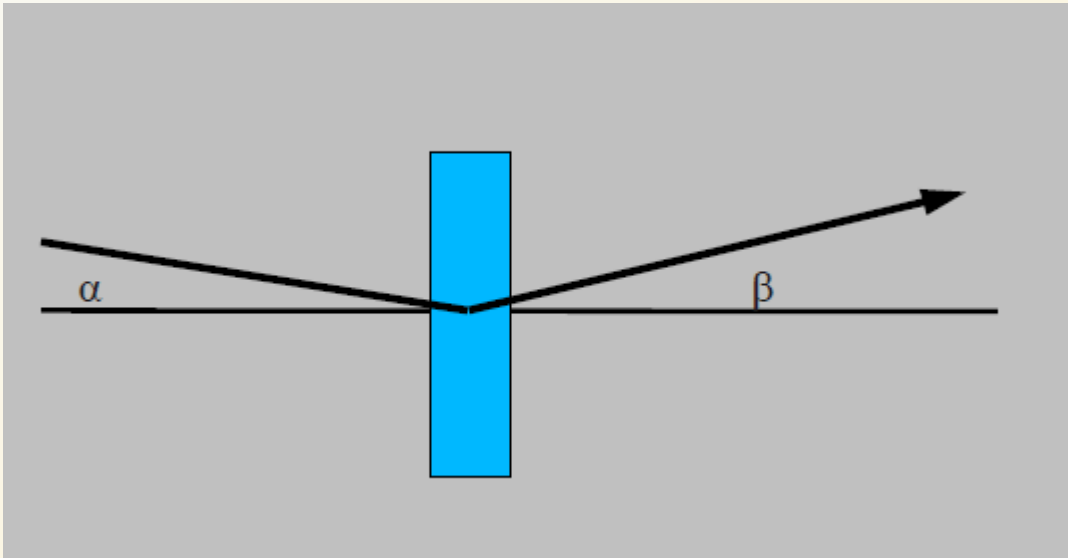
Spectroscopes

- Use either a prism or a diffraction grating.
- ϕ : angle of refraction.
- $dn/d\lambda$: dispersion.
- Grating types: transmission, reflection.
- d : uniform period.
- $1/d$: grating constant.



Spectroscopes: angular dispersion

- Angular dispersion: $d\beta/d\lambda$.
- If the plane wave is incident at angle α relative to the grating normal, in the plane orthogonal to the grating periodicity (d), the grating equation becomes:



$$\sin \alpha + \sin \beta = \frac{m\lambda}{d}$$

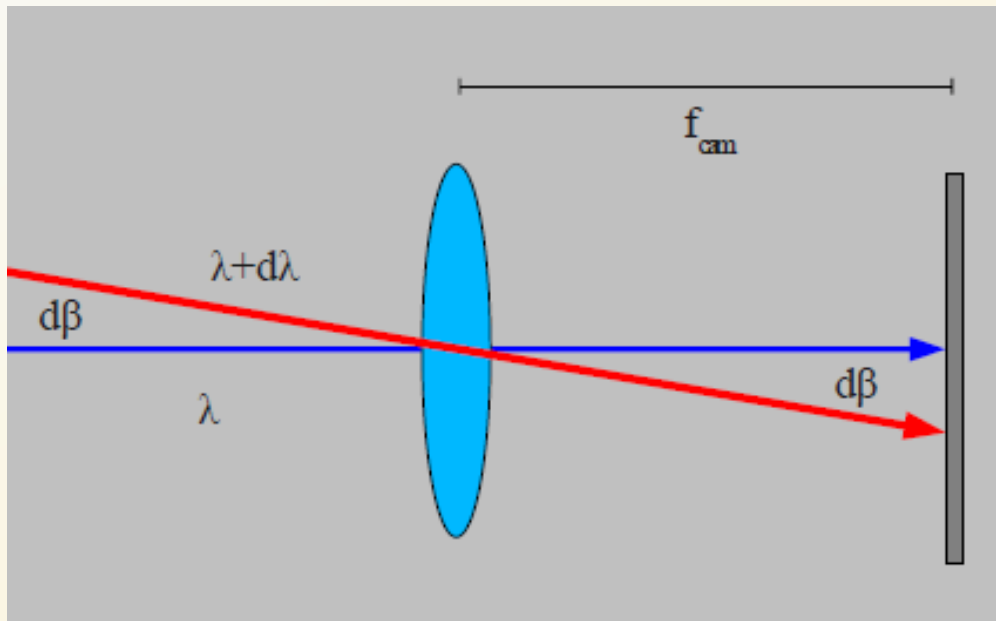
$$\frac{d\beta}{d\lambda} = \frac{m}{d \cos \beta}$$

$$\frac{d\beta}{d\lambda} = \frac{2 \sin \frac{\varphi}{2}}{\cos \frac{\beta + \varphi}{2}}$$

- In case of a prism:
- m is the diffraction order (integer). It describes the angle at which constructive interference occurs when light waves diffract from a grating.

Spectroscopes: linear dispersion

- Linear dispersion: $dx/d\lambda$. It is a measure of the wavelength separation with respect to the change in position of the output spectrum. It quantifies the rate at which different wavelengths are dispersed across the output spectrum.



$$dx = f_{cam} d\beta$$

$$\frac{dx}{d\lambda} = f_{cam} \frac{d\beta}{d\lambda}$$

- Reciprocal linear dispersion:

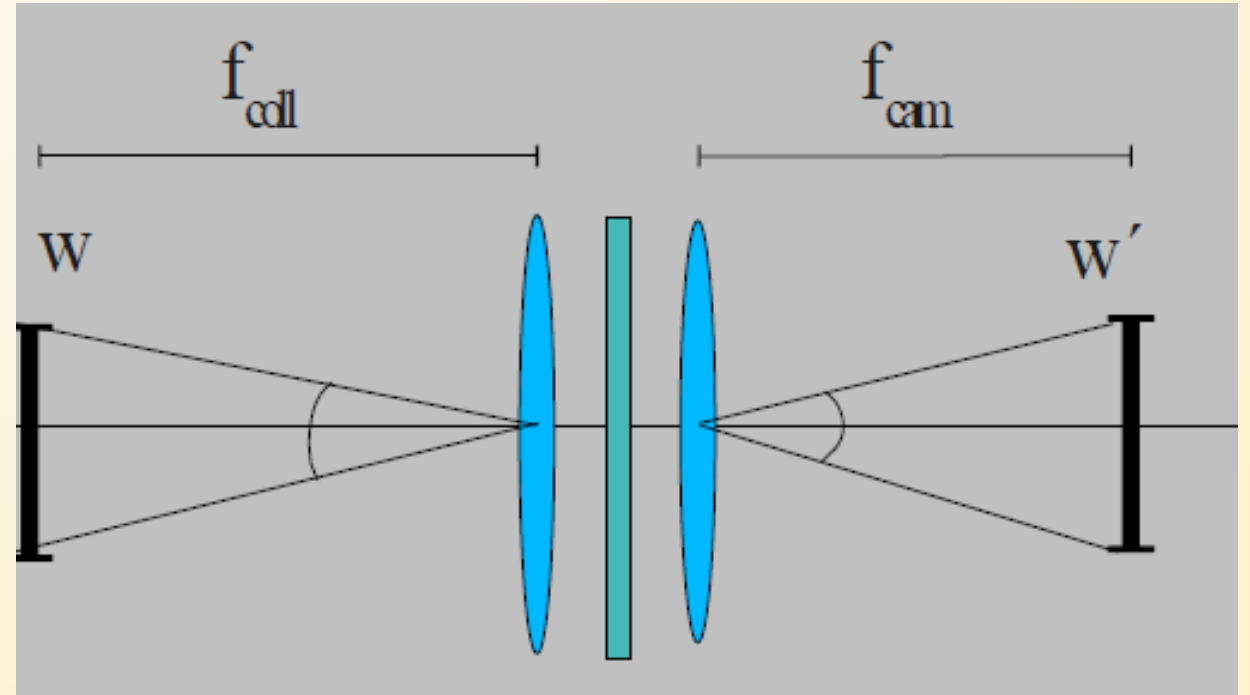
$$\frac{d\lambda}{dx} = \frac{1}{f_{cam} \left(\frac{d\beta}{d\lambda} \right)^{-1}} = \frac{d \cos \beta}{m f_{cam}}$$

Spectroscopes: resolution

$$\frac{w}{f_{coll}} = \frac{w'}{f_{cam}}$$

$$\Delta\lambda = w' \frac{d\lambda}{dx} = w \frac{f_{cam}}{f_{coll}} \frac{d\lambda}{dx}$$

$$\Delta\lambda = \frac{d \cos \beta}{m} \frac{w}{f_{coll}}$$



- Resolution (w is the width of the slit):

$$R = \frac{\lambda}{d\lambda} = \frac{m}{w \cos \beta} \frac{\lambda f_{coll}}{d}$$

Spectroscopes: resolution

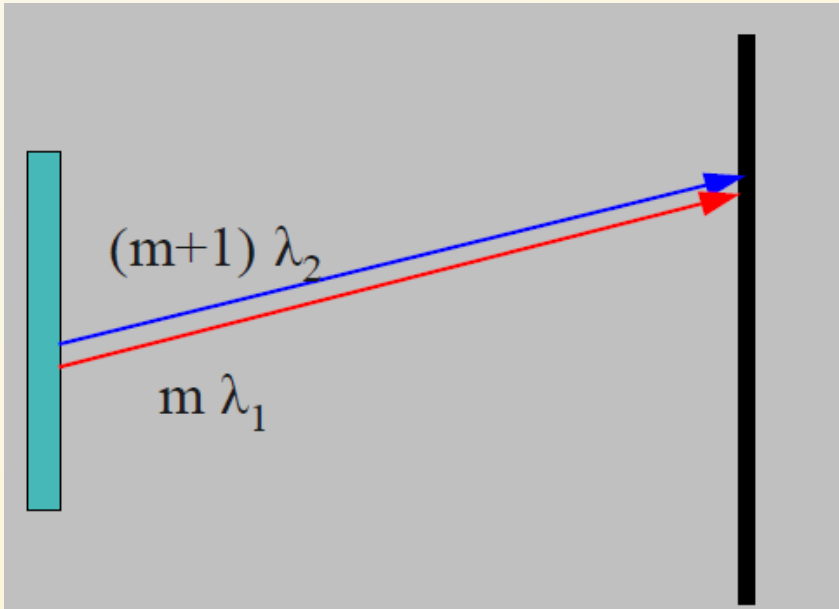
- Prism: $\frac{\lambda}{d\lambda} = b \frac{dn}{d\lambda}$ where b is the base length.
- Grating: $\frac{\lambda}{d\lambda} = W \frac{m}{d}$ where W is the width of the grating.
- The ratio of the spectroscopy and the grating resolution, D is the diameter of the collimator:

$$\frac{R_{sp}}{R_{gr}} = \frac{\lambda f_{coll}}{W w \cos \beta} = \frac{\lambda}{w} \frac{f_{coll}}{D_{coll}}$$

$$\frac{f_{coll}}{D_{coll}} > 1 \quad \frac{\lambda}{w} \ll 1 \quad \text{so} \quad \frac{R_{sp}}{R_{gr}} \ll 1$$

Spectroscopes: resolution and free spectral range

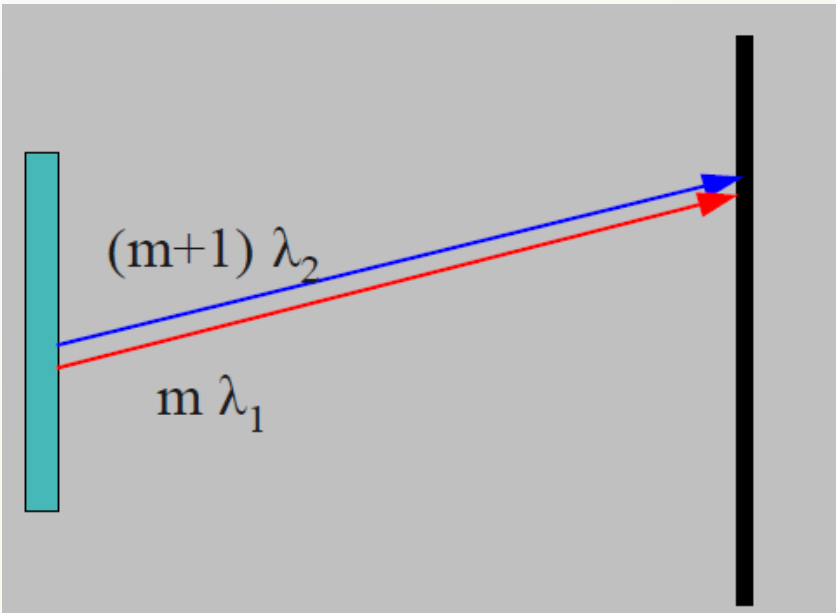
- Low resolution: $R < 1000$
- Medium resolution: $1000 < R < 10\,000$
- High resolution: $R > 10\,000$
- Extreme resolution: $R > 100\,000$
- The free spectral range is the spacing in wavelength between two successive reflected optical intensity maxima or minima of a diffractive optical element.



$$m\lambda_1 = d \sin \beta$$

$$(m+1)\lambda_2 = d \sin \beta$$

Spectroscopes: free spectral range



$$d\lambda = \lambda_1 - \lambda_2$$

$$d\lambda = d \sin \beta \left(\frac{1}{m} - \frac{1}{m+1} \right)$$

$$FSR = d\lambda = \frac{d \sin \beta}{m(m+1)}$$

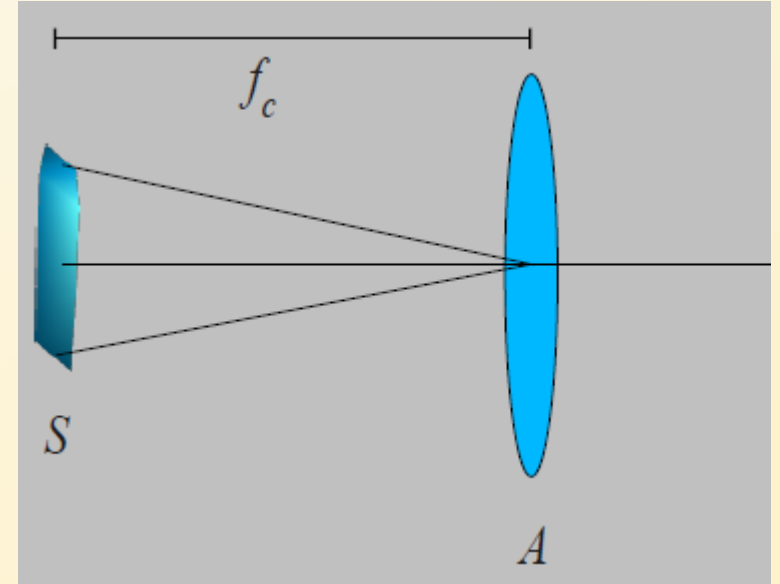
Spectroscopes: luminosity

- Etendue is a property of light in an optical system, which characterizes how spread out the light is in area and angle.

$$S = w l$$

$$A = \frac{\pi}{4 D_{coll}^2}$$

$$\Omega = S/f_c^2$$



- Etendue:
$$E_{sp} = A\Omega = \pi/4 \frac{D_{coll}^2}{f_c^2} w l$$

- Slit lit homogenously: $L = B E$

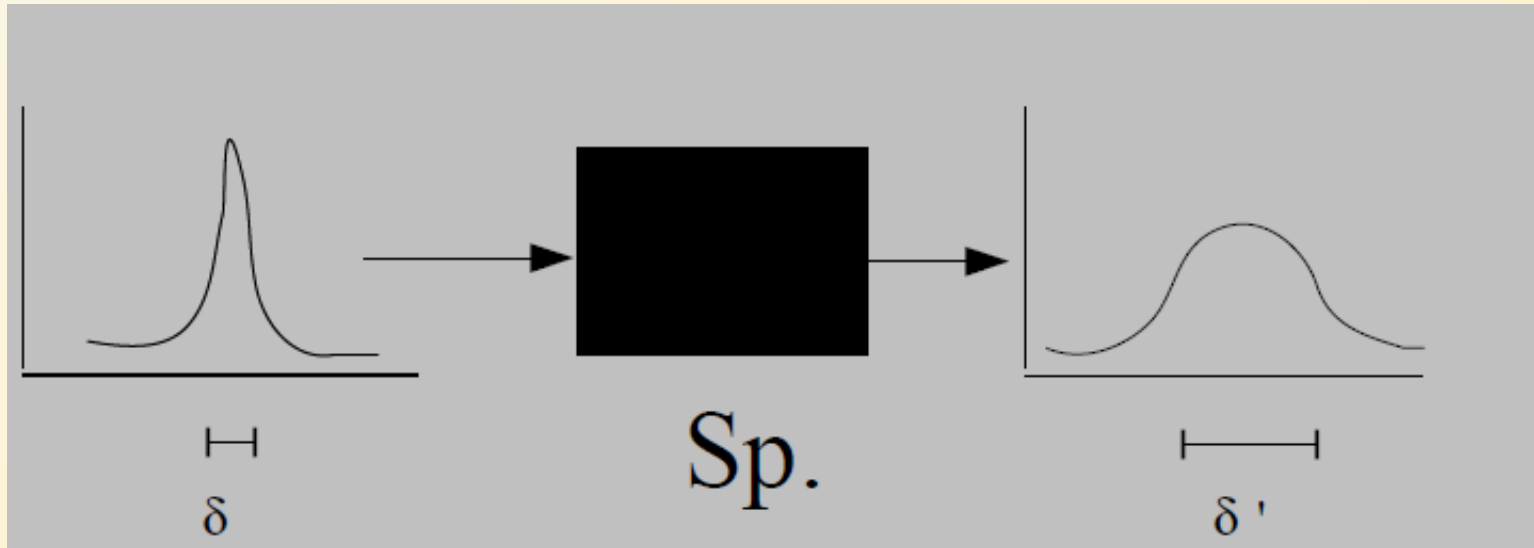
- Optimal lit, σ is the angle diameter of seeing disk:
$$E_{tel} = \frac{\pi}{4 D_{coll}^2} \sigma$$

Spectroscopes: effect of seeing

- Resolution can be increased by decreasing the width of the grating, $R \sim 1/w$.
- If the width of the grating decreases the luminosity does too, $R \sim w$.
- This can be offset by increasing D , thus we need a bigger telescope!
- Typical seeing is $\sigma = 1''$, good seeing is about $0.5''$, Hungary: $2\text{-}3''$.
- The size of the seeing disk on the plane is $s = f_{tel} \sigma$
- If $s > w$ the light entering the slit decreases.

The line spread function

- The image of an infinitely narrow line in the object plane is called the line spread function (LSF).
- The LSF of a spectrograph describes the light distribution at the focal plane as a function of wavelength in response to a monochromatic light source.
- The dominant effect in the observed spectrum is a broadening of the spectral features due to the finite resolution of the instrument.
- LSF can be measured by observations of night sky lines (but avoid the doublets).



The line spread function

- Because the slit has a finite thickness: $\Delta\lambda' > \Delta\lambda$
- The measured spectrum is:

$$f(\lambda) = \int_{-\infty}^{\infty} s(\lambda') \cdot h(\lambda - \lambda') d\lambda'$$

- Where $f(\lambda)$ is the measured spectrum, $s(\lambda')$ is the original spectrum, and $h(\lambda - \lambda')$ is the line spread function.
- The convolution is the multiplication of the Fourier-transforms.
- The original spectrum can be restored with deconvolution, however the noise will significantly increase.

Installation of iraf

Linux in Windows: <https://learn.microsoft.com/en-us/windows/wsl/install>

Iraf: <https://iraf-community.github.io/install.html>

Follow these steps:

1. Start Ubuntu from the Windows Start Menu
2. In the Ubuntu terminal window type *sudo apt get update*
3. The sudo command will ask for your password, so type it in and press enter.
4. Then type *sudo apt install iraf xterm*
5. Now you'll see a whole load of text
6. Press enter to accept the default (capitalized) answer of "Y"
7. Watch as it downloads and installs packages
8. When it's done type *xterm* & and press enter
9. In the xterm window type *irafcl* and press enter
10. In linux you may need to run *mkiraf* first and then run *cl* to start iraf.